

1. Why Do We Need Model Evaluation in Machine Learning?

After building a machine learning model, the next question is:

How good is the model?

A model may produce predictions, but we must check:

- Are predictions correct?
- How large are the errors?
- Can the model generalize to new data?

This process is called **model evaluation**.

Model evaluation measures how well a model performs on **unseen data** using quantitative metrics.

Example

Suppose we built a model to predict **house prices**.

Actual price:

₹90,00,000

Model predicted:

₹88,00,000

Error:

₹2,00,000

Now imagine we test the model on **100 houses**.

Evaluation metrics tell us:

- average error
- prediction accuracy
- reliability of the model

Why Model Evaluation is Important

1. Check Model Accuracy

We must verify whether the model is making correct predictions.

Example:

Spam detection model

If accuracy is **60%**, the model is poor.

2. Compare Multiple Models

Often we build **multiple models**.

Example:

Model Accuracy

Model A 82%

Model B 91%

Clearly:

Model B is better

3. Measure Prediction Errors

Some problems require **minimizing error**, not maximizing accuracy.

Example:

House price prediction.

If the model predicts:

₹90L instead of ₹30L

That is a **huge error**.

Metrics help measure this.

2. Types of Evaluation Metrics

Evaluation metrics depend on **problem type**.

Problem Type	Output	Metrics Used
Regression	Numerical value	MAE, MSE, RMSE, R^2
Classification	Category	Accuracy, Precision, Recall, F1

Classification metrics evaluate categorical predictions, while regression metrics evaluate continuous numerical predictions.

3. Regression Evaluation Metrics

Regression models predict **continuous values**.

Examples:

- house price
- temperature
- sales
- stock price

Metrics measure **difference between predicted and actual values**.

4. Mean Absolute Error (MAE)

Definition

MAE measures the **average absolute difference** between predicted and actual values.

Formula

```
MAE = average(|Actual - Predicted|)
```

Example

Actual Price	Predicted Price
100	90
200	210
300	295

Errors:

```
|100 - 90| = 10  
|200 - 210| = 10  
|300 - 295| = 5
```

MAE:

```
(10 + 10 + 5) / 3 = 8.33
```

Meaning:

On average the model makes ₹8.33 error

Interpretation

MAE Value	Meaning
Small	Better model
Large	Poor model

Python Code

Pseudocode

```
Input actual values  
Input predicted values  
Compute absolute difference  
Take mean
```

Code

```
from sklearn.metrics import mean_absolute_error
```

```
y_true = [100, 200, 300]
y_pred = [90, 210, 295]

mae = mean_absolute_error(y_true, y_pred)

print(mae)
```

5. Mean Squared Error (MSE)

Definition

MSE calculates **average squared error**.

It penalizes **large errors more heavily**.

Formula

$$\text{MSE} = \text{average}((\text{Actual} - \text{Predicted})^2)$$

Example

Actual	Predicted
100	90
200	210

Errors:

$$(100 - 90)^2 = 100$$
$$(200 - 210)^2 = 100$$

MSE:

$$(100 + 100) / 2 = 100$$

Why Square Errors?

Because large mistakes should be penalized more.

Example:

Error:

10 → 100
20 → 400

Large errors grow rapidly.

Python Code

```
from sklearn.metrics import mean_squared_error  
  
mse = mean_squared_error(y_true, y_pred)  
  
print(mse)
```

6. Root Mean Squared Error (RMSE)

Definition

RMSE is the **square root of MSE**.

It converts error back to **original unit**.

Formula

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Example

If:

$$\text{MSE} = 100$$

Then:

$$\text{RMSE} = 10$$

Meaning:

Average prediction error ≈ 10 units

Python Code

```
import numpy as np
from sklearn.metrics import mean_squared_error

rmse = np.sqrt(mean_squared_error(y_true, y_pred))

print(rmse)
```

7. R^2 Score (Coefficient of Determination)

Definition

R^2 measures **how well the model explains the variance in data**.

Range

Value	Meaning
1	Perfect prediction
0	No predictive power
<0	Worse than baseline

Example

If:

$$R^2 = 0.85$$

Meaning:

Model explains 85% variation in data

Python Code

```
from sklearn.metrics import r2_score

r2 = r2_score(y_true, y_pred)
```

```
print(r2)
```

8. Classification Evaluation Metrics

Classification models predict **categories**.

Examples:

- spam vs not spam
 - fraud vs legitimate
 - disease vs healthy
-

9. Confusion Matrix

All classification metrics are derived from **Confusion Matrix**.

It compares **actual vs predicted values**.

Example:

	Predicted Yes	Predicted No
Actual Yes	TP	FN
Actual No	FP	TN

Meaning

TP = True Positive
FP = False Positive
FN = False Negative
TN = True Negative

Example

Disease detection.

Case	Meaning
TP	Sick person correctly detected
TN	Healthy person correctly detected
FP	Healthy person predicted sick
FN	Sick person predicted healthy

10. Accuracy

Definition

Accuracy measures **overall correctness of predictions**.

Formula

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total}$$

Example

Total Predictions Correct

100 90

Accuracy:

$$90 / 100 = 90\%$$

Python Code

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_true, y_pred)

print(accuracy)
```

11. Precision

Definition

Precision measures **how many predicted positives were actually correct.**

Formula

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Example

Spam detection.

Emails flagged as spam	Actually spam
100	80

Precision:

$$80 / 100 = 80\%$$

Meaning:

80% spam predictions were correct

12. Recall (Sensitivity)

Definition

Recall measures **how many actual positives were detected.**

Formula

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Example

Suppose:

Actual spam emails	Detected spam
100	90

Recall:

$$90 / 100 = 90\%$$

Meaning:

Model detects 90% of spam emails

13. F1 Score

Definition

F1 Score balances **Precision and Recall**.

It is the harmonic mean of precision and recall.

Formula

$$F1 = 2 * (Precision * Recall) / (Precision + Recall)$$

Example

Precision = 0.8

Recall = 0.9

$$F1 = 0.847$$

Meaning:

Balanced performance.

Python Code

```
from sklearn.metrics import f1_score
```

```
f1 = f1_score(y_true, y_pred)

print(f1)
```

14. When to Use Which Metric

Scenario	Best Metric
House price prediction	RMSE
Sales prediction	MAE
Balanced classification	Accuracy
Fraud detection	Recall
Spam filtering	Precision
Imbalanced dataset	F1 Score

15. Real Example

Cancer Detection

False negatives are dangerous.

Meaning:

Sick patient predicted healthy

Therefore use:

Recall

Email Spam Detection

False positives are annoying.

Meaning:

Important email marked spam

Therefore use:

Precision

16. Regression vs Classification Metrics Summary

Metric	Used For
MAE	Regression
MSE	Regression
RMSE	Regression
R^2	Regression
Accuracy	Classification
Precision	Classification
Recall	Classification
F1 Score	Classification

17. Simple Intuition Summary

Regression metrics measure:

How far predictions are from actual values

Classification metrics measure:

How many predictions are correct